

Template Week 1 – Bits & Bytes

Student number:

Assignment 1.1: Bits & Bytes intro

What are Bits & Bytes?

- Bit = the smallest piece of data a computer can understand.
It can only be 0 or 1 — like an on/off switch.
- Byte = a group of 8 bits.
Example: 1 Byte = 8 Bits

Computers use bits and bytes to store and process all kinds of information — text, pictures, videos, everything.

What is a nibble?

A nibble is half a byte, consisting of 4 bits. It can represent 16 different values, which is particularly useful in hexadecimal notation where each nibble corresponds to a single hex digit (0-9, A-F).

What relationship does a nibble have with a hexadecimal value?

A nibble consists of 4 bits and can represent 16 different values, ranging from 0 to 15 in decimal. This directly corresponds to a single digit in the hexadecimal (base-16) numbering system, which uses the digits 0–9 and letters A–F to represent values from 0 to 15.

Hexadecimal notation simplifies the representation of binary data by grouping bits into nibbles and expressing each nibble as one hex digit, making it easier to read and write large binary values.

Why is it wise to display binary data as hexadecimal values?

- Compactness:
Binary numbers can be very long and hard to read. Hexadecimal condenses every 4 bits (a nibble) into a single hex digit, making the data much shorter and easier to interpret.
- Readability:
Hexadecimal numbers are simpler to read and write compared to long strings of 0s and 1s. This reduces errors when analyzing or debugging data.
- Direct Mapping:
Since each hex digit corresponds exactly to 4 binary bits, conversion between binary and hexadecimal is straightforward, preserving data accuracy.

- **Standard Usage:**
Hexadecimal is widely used in programming, memory addresses, and digital electronics, making it a common and accepted way to represent binary data.

What kind of relationship does a byte have with a hexadecimal value?

A byte is made up of 8 bits and can represent 256 different values (0 to 255 in decimal). Since each hexadecimal digit corresponds to 4 bits (a nibble), a byte can be represented by two hexadecimal digits.

An IPv4 subnet is 32-bit, show with a calculation why this is the case.

An IPv4 address consists of 4 groups of numbers called *octets*. Each octet is made up of 8 bits. Since there are 4 octets: Total bits = $4 \times 8 = 32$ bits

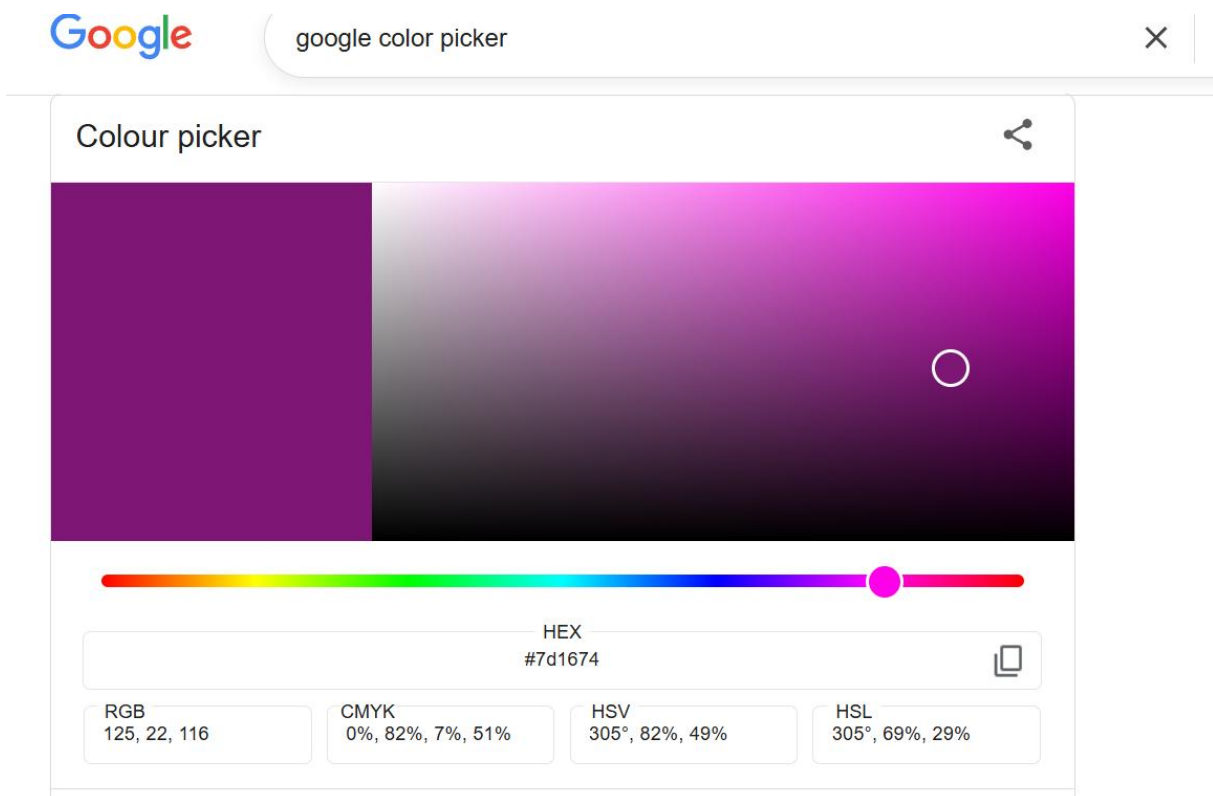
Assignment 1.2: Your favourite color

Hexadecimal color code:

Assignment 1.3: Manipulating binary data

Color	Color code hexadecimaal (RGB)	Big Endian	Little Endian
RED	FF0000	FF0000	0000FF
GREEN	00FF00	00FF00	00FF00
BLUE	0000FF	0000FF	FF0000
WHITE	FFFFFF	FFFFFF	FFFFFF
Favourite (previous assignment)	7D1674	7D1674	74 16 7D

Screenshot modified BMP file in hex editor:



New file

Open file

Save as

Undo

Redo

Tools

Settings

Help

File Information

4pixels2.bmp

134 bytes

Data Inspector (Little-endian)

Unsigned (+)

Signed (±)

Integer

Integer

Integer

Integer

Integer (+)

Integer (±)

Float, P.

Float, P.

Float, P.

End of file

End of file

End of file

End of file

End of file

End of file

End of file

End of file

End of file

4pixels2.bmp

42 4D 86 00 00 00 00 00

00 00 04 00 00 00 01 00

00 00 0C 00 00 00 13 0B

00 00 00 00 00 00 42 47

00 00 00 00 00 00 00 00

00 00 00 00 00 00 00 00

00 00 00 00 00 00 00 00

00 00 00 00 00 00 00 00

00 00 00 00 00 00 00 00

FF 00 00 74 16 7D +

00 00 7A 00 00 00 6C 00

00 00 01 00 18 00 00 00

00 00 13 0B 00 00 00 00

52 73 00 00 00 00 00 00

00 00 00 00 00 00 00 00

00 00 00 00 00 00 00 00

00 00 02 00 00 00 00 00

00 00 00 FF 00 FF 00

BMā.....z...l.

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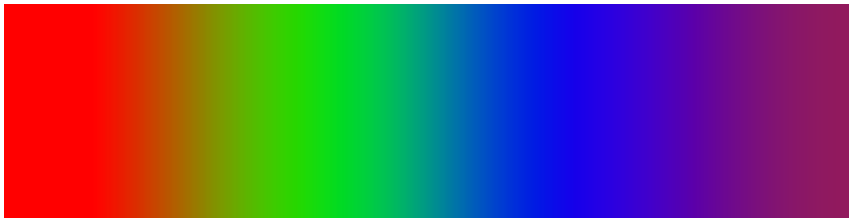
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Assignment 1.4: Student number to HEX and Binary

Convert your student number to a hexadecimal number and a binary number.

556707 is my student number.

Decimal to Hexadecimal

$$556,707 \div 16 = 34,794 \text{ remainder } 3$$

$$34,794 \div 16 = 2,174 \text{ remainder } 10 \text{ (which is A in hex)}$$

$$2,174 \div 16 = 135 \text{ remainder } 14 \text{ (which is E)}$$

$$135 \div 16 = 8 \text{ remainder } 7$$

$$8 \div 16 = 0 \text{ remainder } 8$$

8 7 E A 3 $\rightarrow$ so 556,707 decimal = 87EA3 hexadecimal.

Decimal to binary

The largest power of 2 $\leq$ 556,707 is 524,288 (which is 2^{19}).

$$556,707 - 524,288 = 32,419$$

Since 524,288 fits, the bit for 2^{19} is 1.

1. $2^{18} = 262,144$
the bit for 2^{18} is 0.
2. $2^{17} = 131,072$
bit is 0.
3. $2^{16} = 65,536$
bit is 0.
4. $2^{15} = 32,768$
bit is 0.
5. $2^{14} = 16,384$
bit is 1. Subtract: $32,419 - 16,384 = 16,035$.
6. $2^{13} = 8,192$
bit is 1. Subtract: $16,035 - 8,192 = 7,843$.
7. $2^{12} = 4,096$
bit is 1. Subtract: $7,843 - 4,096 = 3,747$.
8. $2^{11} = 2,048$
bit is 1. Subtract: $3,747 - 2,048 = 1,699$.

9. $2^{10} = 1,024$
bit is 1. Subtract: $1,699 - 1,024 = 675$.
 10. $2^9 = 512$
bit is 1. Subtract: $675 - 512 = 163$.
 11. $2^8 = 256$
bit is 0.
 12. $2^7 = 128$
bit is 1. Subtract: $163 - 128 = 35$.
 13. $2^6 = 64$
bit is 0.
 14. $2^5 = 32$
bit is 1. Subtract: $35 - 32 = 3$.
 15. $2^4 = 16$
bit is 0.
 16. $2^3 = 8$
bit is 0.
 17. $2^2 = 4$
bit is 0.
 18. $2^1 = 2$
bit is 1. Subtract: $3 - 2 = 1$.
 19. $2^0 = 1$
bit is 1. Subtract: $1 - 1 = 0$.
- So the binary number is 1 0 0 0 0 1 1 1 1 0 1 0 1 0 0 0 1 1

Explain in detail that the calculation is correct. Use the PowerPoint slides of week 1.

Ready? Save this file and export it as a pdf file with the name: [week1.pdf](#)